Adam MOUNIR

AI Research / Engineering Student

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EDUCATION

CentraleSupélec - Université Paris-Saclay **Paris, FR**

MSc. - Artificial Intelligence *2026 - 2027*

- **S1 :** Optimization for ML - Decision Modeling - Machine Learning - Deep Learning
- **S2 Electives (MVA - ENS Paris-Saclay) :** Graph ML - Stochastic Optimization - Advanced Deep Learning - NLP - Reinforcement Learning

University of California, Irvine **California, USA**

Exchange Student, MEng. in Computer Science *2023 - 2024*

- **Courses :** Mathematics for Cryptography - Java Programming - Computer Architecture - Network Protocols

EFREI - Paris Panthéon-Assas **Paris, FR**

MEng. - Engineering Degree in Data & Artificial Intelligence *2021 - 2026*

- **Courses :** Convex Optimization - Stats/Probas - Graph Theory - Machine / Deep Learning - Reinforcement Learning - NLP

PUBLICATIONS

EEG/EMG Foundation Challenge 2026: Neural Interfaces for Generalizable Decoding

NeurIPS 2026 *Competition Track (under review) - B. Aristimunha, A. Caillet, …,* ***A. Mounir****, …, S. Chevallier, A. Delorme*

Large-scale competition benchmarking models for cross-subject EEG and EMG decoding, co-organized with Meta and leading academic labs; built the benchmark infrastructure.

Neuron-Level Architecture Growth: A Controlled Evaluation for EEG Time-Series Decoding

IEEE ICASSP 2027 *(under review) -* ***A. Mounir****, S. Douka, A. Caillet, B. Aristimunha, S. Chevallier*

Neuron-level growth for EEG time-series decoders: networks widen during training instead of using a fixed width, reaching +2.9 pp over ShallowFBCSPNet at 0.57× parameters; the refusal rate of growth predicts when it helps.

PROFESSIONAL EXPERIENCE

Inria - TAU (Université Paris-Saclay) **Paris, FR**

AI Research - Intern *May 2026 - Nov 2026*

Researching deep-learning architectures for EEG decoding in Braindecode and Growing Neural Networks during training in EEGrow

- Design transformer, graph-neural-network, convolutional and Riemannian models for neural decoding using open EEG datasets
- Co-organize the NeurIPS 2026 EEG decoding competition (Meta, top academic labs) and building its benchmark infrastructure
- Porting in-training architecture growth (gromo) to EEG decoding: conv-junction width auto-sized during training rather than tuned

Thales **Paris, FR**

AI Research - Intern *Nov 2024 - Apr 2025*

Researched global-scale Root Cause Analysis by distilling multi-step reasoning capabilities from LLMs into specialized architectures

- Designed a supervised alignment pipeline to map expert causal trajectories from logs into internalized reasoning schemas
- Distilled causal traces via GPT-4o to fine-tune a SFT pipeline using Llama3.1 (QLoRA) targeting latent reasoning convergence
- Achieved a 0.64 F1-score in RCA prediction and established a validation protocol for Out-of-Distribution scenarios

Société Générale **Paris, FR**

AI Engineer - Apprentice *Oct 2025 - Apr 2026*

Detected semantic discrepancies between unstructured document disclosures and structured multi-provider data

- Automated cross-source data alignment and causal influence tracing within hierarchical scoring models
- Engineered a Zero-shot NER pipeline using Claude 3.5 Sonnet (RAG) and Directed Graphs for topological impact analysis

OPEN-SOURCE CONTRIBUTIONS

Braindecode

Open-Source Contribution | Github Link

Maintaining and extending the reference PyTorch/skorch library for end-to-end neural time-series decoding

- Implementing transformer, graph-neural-network and Riemannian architectures for EEG/MEG signal decoding
- Building reproducible benchmarking pipelines over open neurophysiology datasets (MOABB integration)
- Contributing tested and documented modules to a community-driven library used across academic and clinical labs

PERSONAL INTERESTS & SKILLS

Hobbies : Running, Swimming, Reading, Piano

Languages: French (native), English (C1), Spanish (B2)

Programming Skills: Python/PyTorch, C/Assembly 68000, Java/C#/Scala, MATLAB, HTML/CSS/JavaScript, Oracle, SQL